

Calculus Refresher

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1 Personal Notes

These notes are a concise collection of derived problems from the examples from Calculus books mentioned in the references.

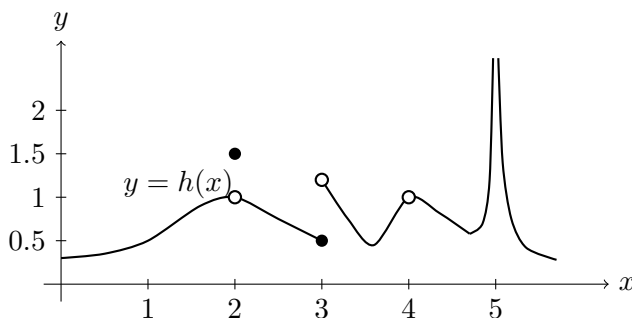
2 Calculus I

2.1 Reading Limits from a Graph

From the graph of $y = h(x)$ below, find $\lim_{x \rightarrow a} h(x)$, if it exists, for

$$a = 1, 2, 3, 4, 5.$$

Also state $h(a)$ whenever it is defined.



Answer

From the graph we read:

$$\lim_{x \rightarrow 1} h(x) = 0.5$$

since the graph is continuous there.

$$\lim_{x \rightarrow 2} h(x) = 1$$

even though

$$h(2) = 1.5.$$

At $x = 3$, the left-hand limit is 0.5 while the right-hand limit is 1.2, so

$$\lim_{x \rightarrow 3} h(x)$$

does not exist.

At $x = 4$, the graph approaches 1 from both sides, so

$$\lim_{x \rightarrow 4} h(x) = 1,$$

but $h(4)$ is not defined.

As x approaches 5, the function grows without bound, so

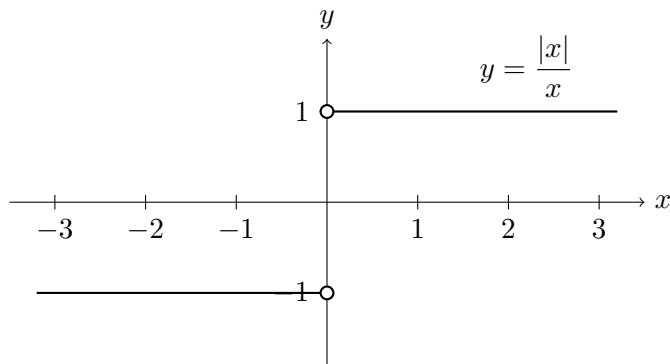
$$\lim_{x \rightarrow 5} h(x)$$

does not exist as a finite number.

2.2 Find $\lim_{x \rightarrow 0} \frac{|x|}{x}$

Answer

This is better visually but we can see that for $x > 0$ the function is 1 and for $x < 0$ the function is -1, and at $x = 0$ the denominator is 0 and the function is undefined at that point. Therefore the limit does not exist at 0.



2.3 Find $\lim_{x \rightarrow \infty} \frac{1}{x}$

Answer

This will be simply 0, since denominator tends to ∞ and the resultant fraction will tend to 0.

2.4 Find $\lim_{x \rightarrow \infty} \frac{7x - 2}{11x + 9}$

Answer

$$\lim_{x \rightarrow \infty} \frac{7x - 2}{11x + 9} = \lim_{x \rightarrow \infty} \frac{7 - 2/x}{11 + 9/x} = \frac{7}{11}$$

2.5 Find $\lim_{x \rightarrow -\infty} \frac{4x^2 - 5x + 3}{x^2 + 2}$

Answer

$$\lim_{x \rightarrow -\infty} \frac{4x^2 - 5x + 3}{x^2 + 2} = \lim_{x \rightarrow -\infty} \frac{4 - 5/x + 3/x^2}{1 + 2/x^2} = 4$$

3 References

Marsden, Jerrold E. and Alan Weinstein, *Calculus I, II, III*.

<https://www.cds.caltech.edu/~marsden/volume/Calculus/>